

- An integrated recruitment platform that combines public job discovery, reviewed CV processing, CV–JD matching, personalized recommendation, application management, Recruiter Jobs, Talent Pool, and a report-moderation workflow.
- A transparent scoring design that keeps the TF-IDF/cosine direct-match score separate from the versioned Potential assessment and explains transferable skills, shared foundations, and seniority checks.
- A feedback-learning workflow based on the Rocchio method, designed to update learned representations from explicit feedback and to support reproducible recomputation rather than uncontrolled cumulative modification.
- A policy-driven AutoFit layer that separates relevance scoring from business actions and applies Human-in-the-Loop controls, user consent, operational constraints, and previous interaction states.
- An actionable email mechanism with expiring, hashed tokens, a non-mutating GET confirmation page, POST redemption, and action-state tracking.
- A traceable evaluation structure that distinguishes algorithm experiments from functional, integration, acceptance, performance, reliability, and security evidence, while explicitly documenting threats to validity.

Thesis objective	Implemented contribution	Evaluation evidence
Analyze IT recruitment workflows	Role-specific requirements and Human-in-the-Loop boundaries	Use cases and traceability matrix
Implement CV–JD matching	TF-IDF, cosine scoring, explanations, and persisted Matchings	Algorithm benchmark and backend tests
Learn from feedback	Rocchio updates followed by scheduled recomputation	Baseline-versus-learned ranking metrics
Support controlled automation	AutoFit policy, quota, cooldown, notification, and audit controls	Security tests and P0 E2E flows
Deliver an integrated platform	Spring Boot, React/TypeScript, PostgreSQL, and Flyway	Build, runtime health, and browser evidence

Table 1.1. Mapping of objectives, contributions, and evidence

7. Thesis Structure

The report includes an Introduction, four chapters covering requirements, design, implementation, and evaluation, and a Conclusion.

PART 2. MAIN CONTENT

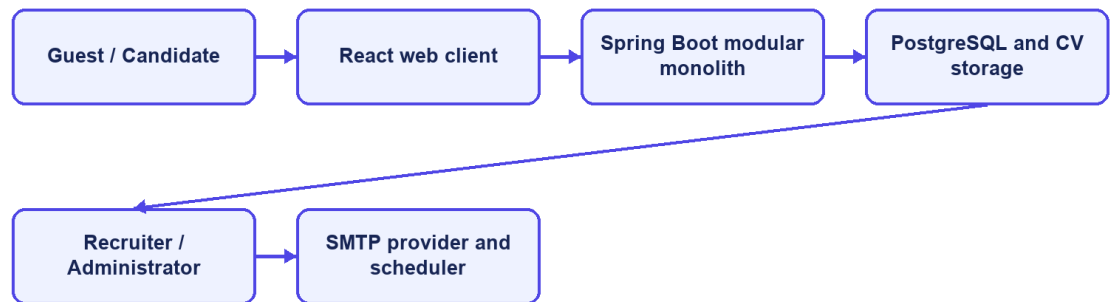
CHAPTER 1. PROBLEM DESCRIPTION AND REQUIREMENTS

1.1 System Analysis Context

CareerFit is designed as a role-based system for IT recruitment. Its boundary includes the React frontend, Spring Boot backend, PostgreSQL database, CV storage, background scheduler, and optional email provider. External job boards, interview scheduling, offers, payroll, and unrestricted automated hiring remain outside the implemented scope.

The central design distinction is between evidence, business state, and action. A matching record contains evidence about the similarity between one CV and one job. An application record represents a candidate's participation in a recruitment process. An automation policy represents consent and operational preferences. These concepts must remain separate: a high similarity score must not silently become an application, and an application status must not be interpreted as a relevance label.

System context



CareerFit IT AutoPilot — thesis diagram

Figure 1.3. CareerFit system context